

Realising the potential of access networks using DSL

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This paper provides an overview of digital subscriber line (DSL) technology and describes some of the key technical issues facing BT's DSL engineering team. The factors which determine the theoretical information-carrying capacity of BT's access network cables are discussed to give the reader a feel for the potential of the existing access network to deliver broadband services. Transmission on this network will be subject to impairments which will limit the achievable capacity, and there is an urgent need to ensure that all DSL systems are spectrally compatible if the capacity of the network is to be fully realised.

1. Introduction

There are around 700 million copper pairs in telephony access networks which provide for extensive connectivity of the world's population. A combination of this existing copper infrastructure and new transmission technologies mean that a new era of universal broadband access can now begin at a fraction of the cost, and in a fraction of the time required for optical access networks.

However, there is a threat to this vision of the future — it is relatively simple to design transmission systems that work well in simulations and in a few specific laboratory tests, but more difficult to deliver useful capacity when subjected to the hostile environment of the real network [1]. Also, the uncontrolled deployment of such advanced transmission systems in multipair cables would result in mutual interference due to crosstalk. Such degradation can occur even when the systems are quite thoroughly specified, since they can be inadvertently operated so as to cause problems. Even a low rate of such instances would significantly pollute the copper network. It is therefore vital to understand the crosstalk environment and spectral compatibility issues for various digital subscriber line variants [2, 3] if the broadband potential of the existing copper access network is to be fully realised.

BT's broadband access strategy involves deployment of ISDN, ADSL and VDSL transmission technologies for a new generation of services, and the use of HDSL for reducing costs and connection time for existing services. BT has therefore invested considerable effort in understanding copper transmission technologies. This is exemplified by high profile involvement in standards bodies such as ETSI, ANSI, ADSL Forum and FSAN. The work has also involved extensive trials, and development of world class test facilities and test methodologies [4].

The issues associated with the spectral compatibility of the various digital transmission systems that have been or will be deployed in the world's copper access networks are technically complex. This paper examines the theoretical information capacity of the copper access network, and explains how measurements have been made on live BT cables to provide realistic performance models. It then describes the critical areas of frequency planning and spectral compatibility which are essential if the broadband potential of this valuable national asset is to be realised.

2. Technology overview

Originally, due to the use of FDM channel multiplexers in the PSTN trunk network, customers were constrained to using the 4 kHz voiceband to convey data over the copper access twisted pair and onwards through the network. Voiceband modems initially worked at just a few tens of bit/s, before progressing through 2.4 and 4.8 kbit/s then developing rapidly to pack more information into the 4 kHz bandwidth. Families of modems operating at 9.6, 19.2, 28.8 and 33 kbit/s subsequently appeared on the market.

With the advent of PCM digital transmission in the core network and digital switching, there was no longer a need to constrain the signals transmitted in the copper access network to a 4 kHz bandwidth. In principle, a copper transmission system could now use any bandwidth as long as the information transported could be conveyed through the 64 kbit/s narrowband digital PSTN.

Data-over-voice pair-gain systems and KiloStreamTM private circuit systems were among the first to make use of additional bandwidth on the copper pairs by using frequencies above the voice-band. Subsequently, basic-rate ISDN systems operating at 160 kbit/s used more modern

modulation schemes (2B1Q, 3B2T and 4B3T), together with advanced digital signal processing, such as nonlinear echo cancellation, to exploit more efficiently the available capacity by improving the spectral efficiency. The transmitted signals for these ISDN systems are mostly constrained to frequencies below 80 kHz.

Next in the evolution of digital transmission systems was HDSL which was originally a 'scaled-up' version of basic rate ISDN. HDSL uses the same line code as the ISDN standard (2B1Q), but operates at 784 kbit/s to 1 Mbit/s on each pair¹. It uses two or three copper pairs to deliver multi-megabit services in the UK, such as ISDN30 and MegaStreamTM.

ADSL is the most recent DSL system to enter commercial deployment. ADSL employs asymmetric transmission in terms of both capacity and bandwidth in order to improve the crosstalk environment, and hence improve the capacity in the direction of exchange to customer compared to HDSL. Unlike ISDN and HDSL which use baseband transmission techniques, ADSL uses passband modulation (between 25 kHz and 1.1 MHz) to keep the 4 kHz voice-band free for simultaneous analogue telephony. Figure 1 illustrates these three basic classes of modern copper transmission systems.

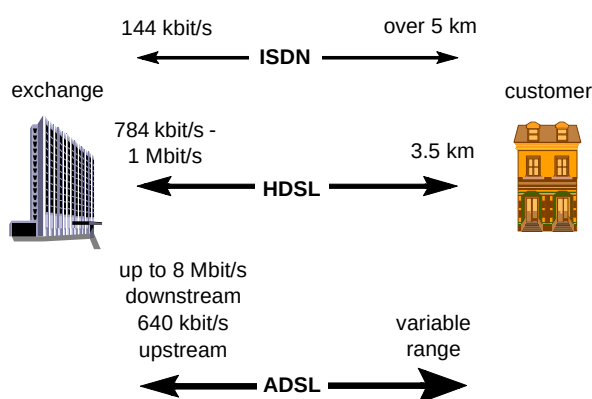


Fig 1 Classes of copper access transmission systems.

All of these systems were designed to operate between the customer's premises and their local exchange. In order to achieve any further significant improvement in capacity, it is necessary to make the copper network shorter. This is achieved by taking fibre part way into the access network and using VDSL for transmission over the remaining D-side and drop-wire copper pairs. VDSL further increased the bandwidth used on the copper pair, with some systems using more than 10 MHz and as high as 30 MHz — see Fig 2. The use of these high frequencies for transmission over copper pairs requires greater attention to electro-magnetic compatibility (EMC) and radio frequency interference (RFI) compatibility during the design [3] because of the diminishing cable balance as the frequency increases.

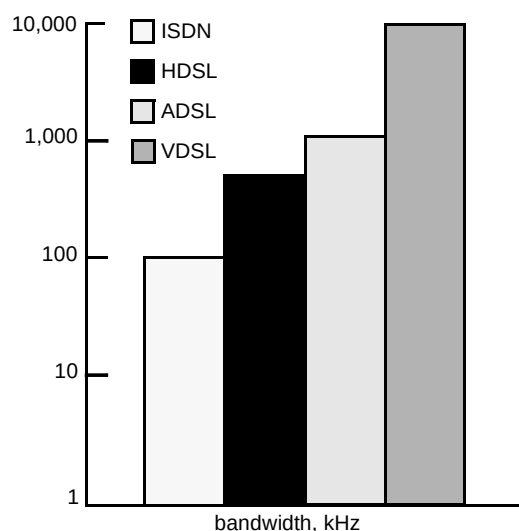


Fig 2 Copper access bandwidth.

Note that it is desirable to use the same line for both the new high rate connections and the original POTS service. This is expected to be achieved by frequency division, using analogue band splitter filters located at each end of the line as shown in Fig 3.

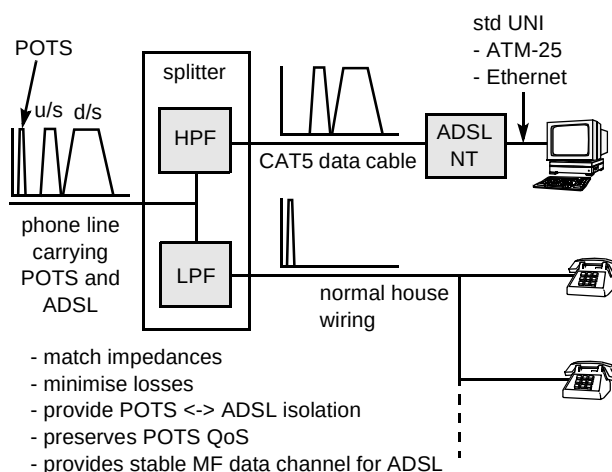


Fig 3 Purpose of a splitter filter².

With the multi-megabit access capability of ADSL and VDSL comes the need to upgrade the remainder of the network to avoid bottle-necks arising elsewhere, e.g. by deploying ATM switches together with SDH and WDM transmission in the core.

Development of new types of DSL modem technology has not halted improvement and development of the existing systems. For example, voice-band modems have evolved to deliver 56 kbit/s to customers by exploiting the reduction in quantisation noise when the analogue/digital conversion process at the POTS line-card is bypassed on the

¹ The bulk of transmitted power being below 400 kHz.

² Normally the high pass filter (HPF) resides in the same box as the ADSL network termination (NT) transceiver and is only shown here collocated with the LPF to illustrate the signal separation.

service provider side. Also, narrowband ISDN line cards with much improved range have become available.

HDSL is no longer seen as just an expedient way of providing existing E1/T1 legacy services using multiple wire-pairs. These systems have become more flexible, e.g. rate-configurable, trading range for bit rate. This potentially opens up new markets for HDSL, such as campus LANs using a single copper pair.

ADSL has also become rate-adaptive, and the systems are no longer simple 'bit pumps' with clock and data interfaces [6]. ADSL systems now exist that have integrated ATM functionality, and even integrated bridging or routing.

Conventionally the ADSL modem is installed close to the master telephone socket, as shown in Fig 4. A recent proposal is to omit the splitter at the customer end. The idea is to remove the need for new internal wiring by allowing the customer ADSL unit to be directly plugged into any existing telephone socket as shown in Fig 5. It would also save the cost of an engineering visit. This is commonly called Splitterless ADSL or ADSL Lite.

For Splitterless ADSL to operate successfully, modifications are needed to the ADSL modem design to compensate for the increased mutual interference between POTS and ADSL. This interference results in additional noise for telephony, and reduced capacity for the ADSL. Moreover, the performance of splitterless ADSL will be very dependent on the type and number of telephones and other telephony CPE that are installed. It may be that, unlike normal ADSL, simultaneous operation of telephony and splitterless ADSL will not be possible for some customers without the addition of a second line (or of course a splitter).

VDSL developments which have arisen since it was first conceived include increased interest in symmetrical transmission, and the use of VDSL from the exchange and in campus environments.

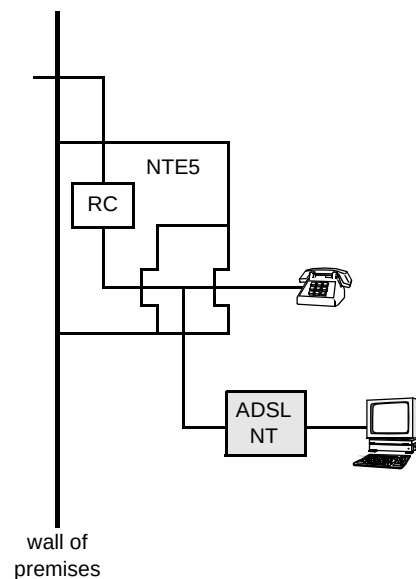


Fig 5 'Splitterless' installation.

All DSL technologies have benefited from the continuing advances in electronics. Apart from improvements in functionality, there is modest scope for DSL modems to access more of the intrinsic information capacity of the copper pairs. This relies on more sophisticated modulation and coding techniques, and making use of improvements in silicon integration to generate improvements at acceptable cost, and reduced power consumption. Table 1 (reproduced from Chen [7])

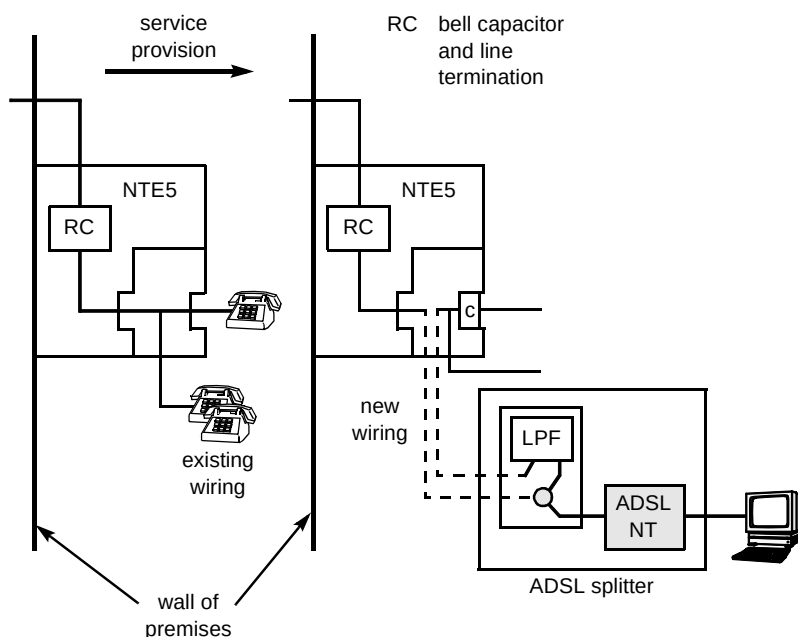


Fig 4 Normal ADSL installation.

compares the capacity actually obtained by some existing systems against the theoretical capacity³ of their channels.

Table 1 Practical capacity as a percentage of theoretical.

Attribute	V.34 modem	ISDN	HDSL	ADSL
Channel capacity (kbit/s)	34.88	700	1700	10 000
Throughput (kbit/s)	28.8 ⁴	160	800	7000
Transmission efficiency (%)	83	23	47	70

The signal processing used in voice-band modems operates at relatively slow rates, enabling sophisticated modulation and coding algorithms, such as multi-dimensional trellis coding, to be implemented at low cost. Voice-band modems are close to the limit in realising the capacity available in the voice band.

Similar techniques were not practical when ISDN and HDSL were originally conceived and standardised, so these systems are less efficient. By the time ADSL appeared, signal processing power and integration had improved and ADSL began to make better use of available capacity than its DSL predecessors by exploiting techniques such as multi-tone modulation, Reed-Solomon error-correction coding, and trellis coding.

The increased availability of fast signal processing power is now being used for new improved variants of the earlier DSL systems. HDSL2 is a new variant of HDSL that seeks to deliver 1.5 Mbit/s T1 services over a single copper pair with similar range to that currently achieved with 2-pair T1 HDSL systems. To achieve this increased transmission efficiency, HDSL2 uses a modulation technique known as POET [8] which is a much more sophisticated approach than the existing 2B1Q line code. It also employs coding techniques.

In addition to improvements in DSL technology, any further improvement in capacity depends on maintaining the available network capacity by judicious control of the crosstalk environment, and spectrum management. For example, HDSL2 has been defined for standardisation in the T1 HDSL market (primarily North America). However, HDSL2 may not be able to deliver 2 Mbit/s E1 services over a single copper pair without causing interference to ADSL systems operating in the same cable.

³ For system details and assumed channel characteristics, see Chen [7].

⁴ On the basis of equivalent assumptions, a 33 kbit/s modem operates at 95% transmission efficiency. 56 kbit/s modems effectively operate over a different channel with higher capacity due to the removal of a source of quantisation noise.

3. Impairments for DSL

Copper access transmission systems face a variety of impairments that present barriers to their operation. These can be broadly classified as intrinsic to the cable environment or extrinsic to the cable.

3.1 Noise

Examples of intrinsic noise impairments are thermal noise, echoes and reflections, attenuation and crosstalk. There are also other components that reside in the cable infrastructure that can impair the operation of DSL systems. These include surge protectors, RFI filters, and in some networks, bridged taps and loading coils.

Another intrinsic impairment is the condition of the cable infrastructure which exhibits faults such as split pairs, bunched pairs, leakage to ground, low insulation resistance, battery or earth contacts, high-resistance joints. All these impairments reduce DSL performance.

Examples of extrinsic impairments are impulsive noise originating from lightning strikes, electric fences, power lines, machinery, arc welders, switches, fluorescent lighting, etc. There is also radio interference from broadcasting and radio transmitters.

The noise sources mentioned above can alternatively be classified as capacity limiting or performance limiting. Capacity-limiting noise is usually slow changing, such as thermal noise and crosstalk. These noise levels are often predictable and relatively easy to take into account when the telco creates deployment-planning rules.

Performance limiting noise, such as impulses and RFI, is intermittent in nature. It is geographically variable and unpredictable, and so is usually accounted for in planning rules by using a safety margin. DSL systems seek to use additional signal processing such as error correction with interleaving, and adaptive line codes, to mitigate such sources of noise.

3.2 Crosstalk

Crosstalk causes by far the largest contribution to capacity limiting noise for DSL systems, so it is worth examining it in a little more detail here. There are two very different types of crosstalk in multi-pair access network cables, near-end crosstalk (NEXT) and far-end crosstalk (FEXT) as shown in Figs 6 and 7.

NEXT is interference that appears on another pair at the same end of the cable as the source of the interference. Its level is substantially independent of the length of the cable.

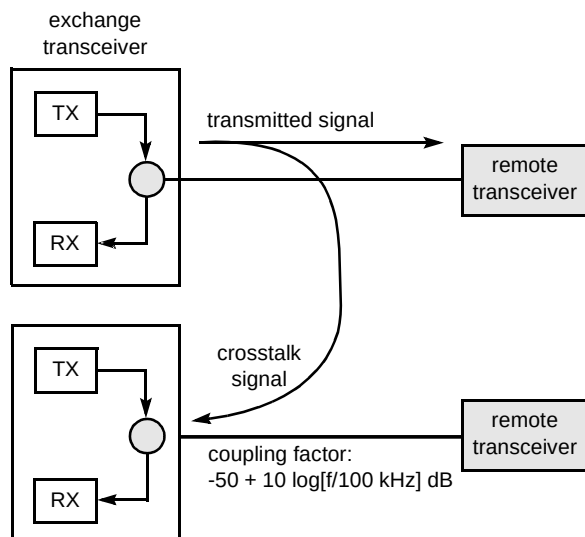


Fig 6 NEXT.

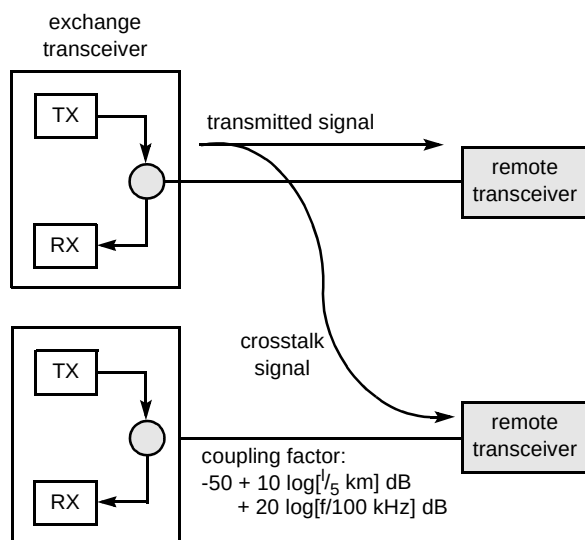


Fig 7 FEXT.

FEXT on the other hand is interference that appears on another pair at the opposite or far end of the cable to the source of the interference. Its level is attenuated at least as much as the signal itself if both have travelled the same distance.

NEXT affects any systems which transmit in both directions at once (e.g. echo-cancelling systems), and where it occurs it invariably dominates over FEXT.

NEXT can in principle be eliminated by not transmitting in both directions in the same band at the same time, separating the two directions of transmission either into non-overlapping intervals in time or into non-overlapping frequency bands (see section 6.2). This converts duplex transmission into independent simplex transmissions, avoiding NEXT at the cost of a reduced bandwidth in each direction. At high enough frequencies the advantage of transmitting against FEXT noise rather than NEXT noise

becomes so great that it can outweigh the disadvantage of reduced bandwidth.

As a rule of thumb the transition between these two modes of transmission occurs at a reach-dependent frequency given by:

$$F_{critical} = \{2.5/(\text{reach}/\text{km}) - 0.45\} \text{ MHz}$$

This compares with heritage systems as shown in Table 2.

Table 2 System reach.

System reach/km	Rule of thumb recommendation	Actual systems
5 km	Multiplexed above 50 kHz	ISDN echo cancelling < 40 kHz ADSL echo cancelling < 138 kHz Multiplexed > 138 kHz
3 km	Multiplexed above 380 kHz	HDSL echo cancelling < 292 kHz
1.5 km	Multiplexed above 1.2 MHz	Fibre to the cabinet VDSL, multiplexed ~1.1—10 MHz

It can be seen that BT's heritage systems are, in the main, optimal according to this rule of thumb.

3.3 EMC

DSL transmission systems are required to operate on access wire-pairs which exist in a harsh physical and electromagnetic environment. As they cannot be screened and are often hung from poles, they have the capacity to act as antennae. This means that they can pick up radiated emissions that may become sources of interference to DSL systems, and, equally, there is the potential for DSL line signals to leak out of the cables and cause interference to radio systems. Obviously it is vital that both of these possibilities are understood and their impact controlled.

Emissions

Most DSL line signals up to and including ADSL use frequencies and signal levels that are so low that emissions are unlikely to radiate significantly. Access network cables transmit DSL signals in a balanced mode (equal and opposite voltages on each wire) that tends to cancel out potential emissions. Any signal that does find its way into the radiative unbalanced mode is likely to be poorly radiated because the wavelength is so long that the antenna efficiency of the cables is very low. This has been

confirmed by measurements both by BT [3] and independent authorities.

At VDSL frequencies the picture changes somewhat. Although the signal levels are still very low, the frequencies are much higher to the extent that the degrading balance of the cables allows more of it to enter the radiative mode. Once there, the shorter wavelength raises the antenna efficiency so that significant emission becomes a more real prospect. By making some careful design choices, the problems can be reduced to manageable proportions however, as has been confirmed by extensive measurements [9–11].

Susceptibility

Antennas work reciprocally. Cables that can radiate emissions can also pick them up from external sources. Normal radio signals do not pose much of a threat to DSL signals though, as the sophisticated receivers used have intrinsic abilities to eliminate or ignore them. RFI pick-up only really becomes an issue when there are strong nearby transmitters. Again VDSL systems are most likely to be affected because of the increased antenna efficiency of network cables at these frequencies. Some special methods can be used to mitigate these effects, such as RFI cancellation. In very severe RFI environments, such as close proximity to a strong AM broadcast transmitter, DSL systems may not be workable and substitute technology, such as direct fibre, may be the only alternative.

4. Measuring FEXT

FEXT is a well-understood problem for telephony cables at low frequencies. It is, however, not well characterised for higher frequencies in a real network. In a real network, FEXT is not just a function of the crosstalk in the cables, but also a function of joints, gauge changes, etc. In order to understand the implications for VDSL, equipment has been developed to measure FEXT in a live network.

Measuring FEXT is both time consuming and involved even in a laboratory environment; in the access network it is even more problematic. In a live network not only is it more difficult to make any measurement, but also the disruption to the network has to be kept to a minimum.

In the laboratory, though cumbersome, it is straightforward to measure the FEXT in a cable. Typically the cable is either new or well maintained with few faults; also the relative arrangement of the pairs in the cable is known. Because FEXT between two pairs drops off markedly with their separation in a cable, and the couplings between two adjacent pairs will be similar from pair to pair, the FEXT SNRs do not have to be measured for all pair combinations to have confidence in the characterisation of

the cable. This, though, is not the case in the access network, and to adequately characterise a cable here, it is necessary to measure as large a set of the pair-to-pair combinations as possible.

Figure 8 is a schematic of part of a distribution network. There are two major differences from the simple cable case:

- there are joints (represented by 'X' in Fig 8),
- sets of pairs terminate at distribution points (DPs), resulting in different lengths over which crosstalk can affect the signal.

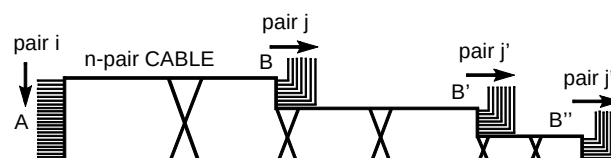


Fig 8 Schematic of a distribution network.

In order to measure FEXT in a live network, VDSL splitters are spliced into all 100 pairs at the cabinet; similarly VDSL splitters are spliced into all the pairs at a given DP (see Fig 9). This reduces the disruption to the customer's POTS service to the time it takes to splice in the splitter — a few seconds — and not the time it takes for the measurement, which is in the order of an hour.

A test signal generator is constructed which generates ten separate, distinct and broadband output signals. These signals are connected to up to ten of the VDSL connections at the DP. A scanning measurement then samples the signals on each of the up to 100 VDSL connections at the cabinet sequentially. As it measures each pair it is able to separate contributions to that pair from each of the up to ten connected distinct transmitters, so measuring up to ten FEXT couplings at up to 100 frequency points.

This is repeated at all the DPs present, thereby measuring all the required FEXT couplings to adequately characterise that cable.

The data is then analysed to extract the complete coupling matrix for the cable being measured of size n by m by l where n is the number of pairs scanned, m is the number of those pairs which had one of the transmitters connected at the DP at some stage during the testing, and l the number of different frequency points measured. The cable attenuations are accounted for, and the matrix can then be used to measure predictions for FEXT.

Figure 10 shows over-plotted the 100 FEXT SNRs for the 100 pairs of a calibration check cable. Each SNR represents the difference between a particular insertion loss

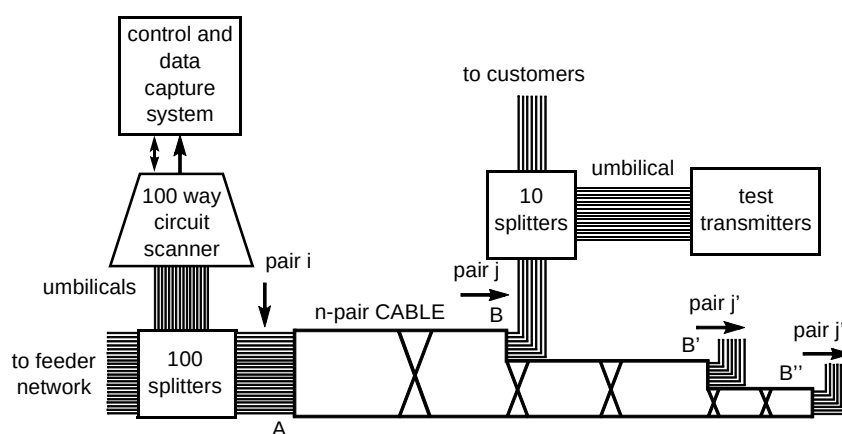


Fig 9 FEXT test arrangement for pairs at DP No B.

and the power sum of the 99 FEXT couplings which contribute to FEXT interference in that pair.

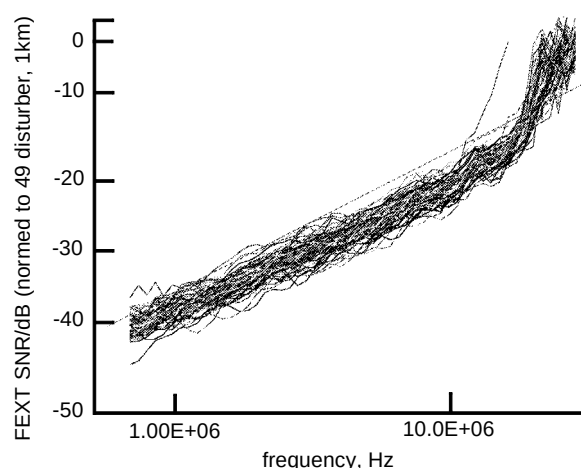


Fig 10 100 FEXT SNRs for LNEL 400 m of 0.4 mm cable.

The straight line in Fig 10 shows the 1% worst-case theoretical model which the work on FEXT is intended to verify or modify. This work will be reported more thoroughly in a future paper.

5. Theoretical capacity

The Hartley-Shannon theorem [12] states that any channel has an intrinsic information capacity⁵, C , for carrying data without error. For a white band-limited Gaussian channel [13] it is given by:

$$C = B \cdot \log_2 \cdot \left(1 + \frac{S}{N}\right) \text{ bit/s}$$

where B is the bandwidth, and S/N is the signal-to-noise power ratio (SNR) at the receiving station.

⁵ This is a mathematical limit. Modern DSL modems achieve performance within a few dB of the limit, but become more complicated and introduce more delay as the limit is approached. At the limit they would be infinitely expensive and never deliver any data because of the processing latency.

For the more general case where signal and noise vary with frequency this can be simply re-arranged thus:

$$C = \int_{f_{min}}^{f_{max}} \log_2 \cdot \left(1 + \frac{S(f)}{N(f)} \right) df \quad \text{bit/s}$$

This expression is classical, dating from the 1940s. However, it is worth re-examining its consequences for DSL applications with the aid of a modern model of our channel [14].

5.1 *NEXT-limited capacity*

For self-NEXT-limited capacity the Hartley-Shannon integral becomes:

$$C = \int_{f_{\min}}^{f_{\max}} \log_2 \cdot \left(1 + \frac{P_t(f) \cdot |H(f)|^2}{P_t(f) \cdot X_n(f)} \right) df \quad \text{bit/s}$$

where $H(f)$ is the voltage transfer function of the channel, $X_n(f)$ is the NEXT power coupling function and $P_t(f)$ is the power spectral density of the transmitter. In computing the self-NEXT-limited capacity some key assumptions must be made.

Firstly, the source of noise in the channel is assumed to be generated by NEXT from adjacent and like systems (hence the term self-NEXT⁶).

The term ‘like system’ is used to infer that the noise sources have the same spectral occupancy or power spectral density (PSD) as the system being interfered with. This is most likely to cause worst-case interference because all the coupled signals will be in-band for the receiver of the interfered system. Also, the worst case will occur when 49 other disturbers in a 50 pair unit⁷ are active at one time.

⁶ Throughout this paper, NEXT limited capacity will imply self-NEXT-limited capacity.

The UK term is *unit*, but the American term *binder group* is more common in the world literature.

Within the unit the individual pairs are usually ordered and twisted in a particular fashion. When a wire-pair suffers interference from other wire-pairs in the same unit it is the nearest 6—8 interferers which cause most of the noise power or crosstalk. However, it cannot be predicted which ones these will be unless *a priori* knowledge of the physical location and spectrum of all the possible interferers in the cable is available. Hence, a worst-case assumption is usually made in which it is assumed that all 49 possible disturbers are active and that 1% worst-case coupling is achieved. This is rather pessimistic but yields a planning rule which is conservative and offers margin to cater for the effects of other (less tangible) noise sources.

Secondly, it must be assumed that the receiver is perfect in all other aspects, namely:

- no nonlinear behaviour in the channel, transmitter or receiver,
- no internal sources of noise (in particular no ADC quantisation noise),
- equalisation cancels all inter-symbol interference (ISI) perfectly,
- echo cancellation removes own-system echoes perfectly.

With the caveats above the expression for the NEXT-limited capacity reduces to:

$$C = \int_{f_{min}}^{f_{max}} \log_2 \left(1 + \frac{|H(f)|^2}{X_n(f)} \right) df \quad \text{bit/s}$$

The expression for NEXT-limited capacity is independent of the transmitter power spectral density and depends only on the channel and crosstalk power coupling characteristics.

In BT's access network two types of metallic twisted pair predominate — 0.4 mm and 0.5 mm diameter copper. Computer simulations of the information capacity were made of lines consisting of one homogeneous length of either 0.4 mm or 0.5 mm cables. Figures 11 and 12 show their NEXT-limited information capacity. With reference to Fig 11, a 3-D plot of NEXT-limited capacity for 0.4 mm⁸ copper cable is shown on the left, whereas a contour plot of the same data is shown on the right.

The 3-D plot shows the surface of maximum capacity (c in Mbit/s) with variation in cable length (l in km) and bandwidth in use from DC to (f_{max} in MHz). Clearly it can be seen that as length increases the capacity decreases. At long cable lengths (~2—5 km), increasing the frequency of transmission has little or no effect on capacity. Only over relatively short lengths is it advantageous to use higher frequencies. Intuitively, this is to be expected from the band-limited nature of metallic access channels. This is illustrated more clearly by the 2-D contour plot shown on the right of the figure. Contours of fixed capacity are plotted for the following capacities — 10, 8, 6, 4, 2 and 1 Mbit/s. The 10 Mbit/s contour is shown at the top left of the plot, whereas the 1 Mbit/s contour is on the right. The 1 Mbit/s contour shows quite clearly that there is no advantage to using frequencies higher than about 200 kHz on long loops⁹. However, the potential for high-speed transmission over short distances, using higher frequencies, would seem to exist.

Figures 11 and 12 show the increase in NEXT-limited capacity with increase in cable gauge. These plots show the remarkably high capacities on offer, and the potential for further exploitation of the local loop by systems occupying bandwidths from DC up to approximately 200 kHz.

In summary, the following conclusions may be drawn from the theoretical work on self-NEXT capacity. More detailed treatment is given in Foster [15]:

⁸ Conductor diameter.

⁹ This assumes a NEXT-limited environment. ADSL avoids this limit by reserving the higher frequencies (up to 1.1 MHz) for one direction only.

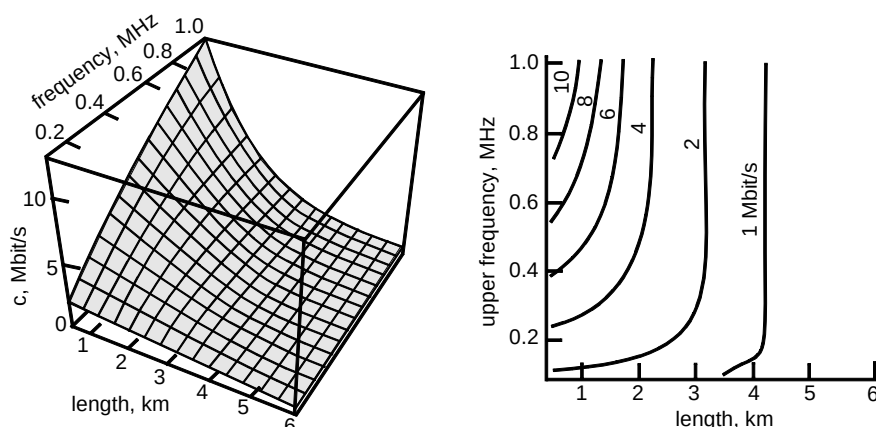


Fig 11 NEXT-limited capacity for 0.4 mm copper cable.

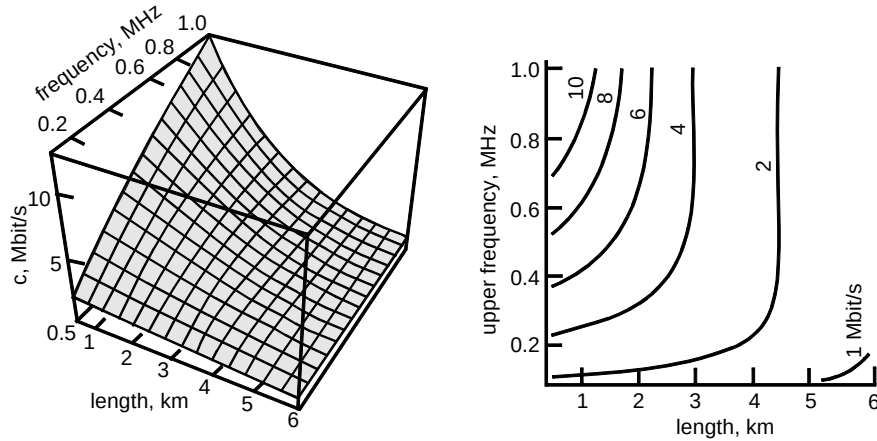


Fig 12 NEXT-limited capacity for 0.5 mm copper cable.

- capacity decreases with increasing cable length,
- for typical access network connections (2—5 km from serving exchange to customer termination), there is little point using frequencies above approximately 200 kHz in a NEXT-limited regime because the increase in capacity is very small,
- there is potential for very high speed transmission (e.g. 25—50 Mbit/s) over relatively short distances — copper tails on the end of passive optical networks may be viable utilising bandwidths up to 10—20 MHz.

5.2 FEXT-limited capacity

For FEXT-limited capacity the Hartley-Shannon integral becomes:

$$C = \int_{f_{\min}}^{f_{\max}} \log_2 \cdot \left(1 + \frac{P_t(f) \cdot |H(f)|^2}{P_t(f) \cdot X_f(f)} \right) df \quad \text{bit/s}$$

where $H(f)$ is the voltage transfer function of the channel, $X_f(f)$ is the FEXT power coupling function and $P_t(f)$ is the power spectral density of the transmitter.

Assuming that the sources of FEXT are due to the FEXT from 49 other disturbers in the 50-pair binder group, and that the receiver is otherwise perfect (as assumed in the self-NEXT-limited analysis), then the above expression reduces to:

$$C = \int_{f_{\min}}^{f_{\max}} \log_2 \cdot \left(1 + \frac{|H(f)|^2}{X_f(f)} \right) df \quad \text{bit/s}$$

Noting that $X_f(f)$ is the accepted Werner FEXT coupling function [16], this further reduces to:

$$C = \int_{f_{\min}}^{f_{\max}} \log_2 \cdot \left(1 + \frac{1}{\psi \cdot l \cdot f^2} \right) df \quad \text{bit/s}$$

where l is the physical length of the channel (in feet), f is frequency (in Hz) and the constant ψ is determined empirically. This is the expression for FEXT-limited capacity. Notice that the expression is independent of transmitted power spectral density, and is frequency and length dependent. It is somewhat surprising that the only dependency on the channel is length, and therefore the FEXT limited capacity will not be dependent on cable type.

Figure 13 shows the FEXT-limited capacity of the access network. On the left is a 3-D plot of the FEXT-limited capacity whereas on the right is a 2-D contour plot of the same data.

Contours of fixed capacity are plotted for the following capacities — 10, 8, 6, 4 and 2 Mbit/s. The 10 Mbit/s contour is shown at the top of the plot whereas the 2 Mbit/s contour is shown at the bottom. It is instructive to compare this figure with a NEXT-limited equivalent (e.g. Fig 11 or Fig 12). Clearly, it can be seen that the contours of constant FEXT capacity are almost orthogonal to the contours of NEXT capacity. Also FEXT would not appear to limit the information-carrying capacity of the channel as much as NEXT does.

In contrast to the NEXT-limited scenario there does appear to be advantage in increasing the frequency components of transmission (i.e. using greater bandwidths will release greater capacity in the FEXT-limited case, as exploited by ADSL and VDSL).

6. Spectral compatibility

Spectral compatibility is concerned with co-ordinating the behaviour of systems sharing the cable environment, in order to realise the intrinsic data transmission capacity of the cables. Earlier sections have discussed mutual interference between the systems as the dominant noise source. Spectral compatibility is achieved by limiting the spectral power each transmitter may inject into the cables.

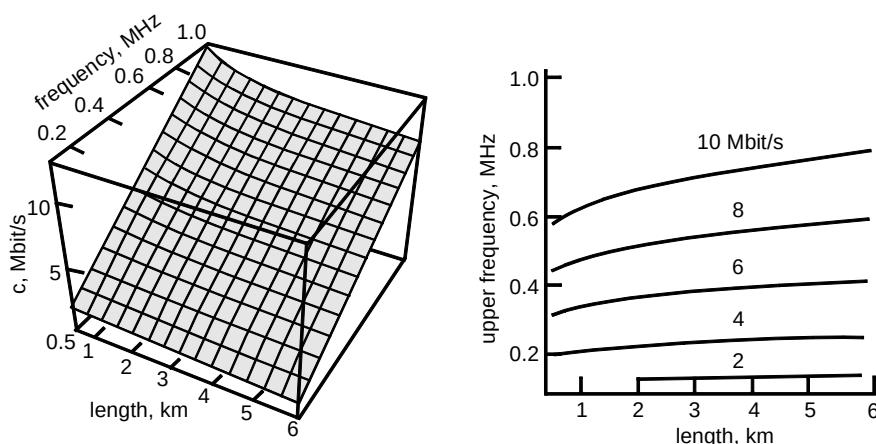


Fig 13 FEXT-limited capacity.

The set of rules which defines these limits is called a ‘frequency plan’.

The frequency plan described here is based on spectral masks [17]; under such a plan, every transmitter has a mask applicable to it, and conforms to the plan providing that its power output is no greater than the mask at each frequency. Which mask is applicable depends on where the transmitter is located, but is independent of the type of the system — the spirit of such plans is that if any type of system may transmit at a given level at a given location, then every type of system may.

6.1 Performance

The performance of a transmission system can either be evaluated in terms of the best data rate it can achieve on a given channel, or the worst channel it can use for a given data rate. The former is more usual for comparing voiceband modems; the latter is more useful to telcos, as it relates to ‘reach’ — the longest line a given technology can use.

When a system uses the same frequency band in both directions of transmission at the same time, adjacent to a system just like itself in the same cable, the system performance is NEXT-limited. As noted previously, both narrowband ISDN and HDSL operate in this way.

The comparisons of Table 1 show ISDN and HDSL achieving a fraction of their (max reach) channels’ capacities, and ADSL achieving a larger fraction. ADSL is less mature, and the difference is a reflection on the improvements to modulation technology. However, even if ISDN and HDSL were re-engineered with newer modulation methods, the gain in reach would be small — this can be seen in the curves in Figs 11 and 12.

From the inception of ADSL, it was clear that to go further required systematic avoidance of NEXT — the

original ADSL proposal was effectively unidirectional, using telephone key tones as the reverse channel [18].

6.2 Duplexing

Systematic separation of the directional channels is called duplexing. Duplexing can be achieved either by the two directions¹⁰ being assigned non-overlapping frequency bands, called frequency division duplexing (FDD), or by assigning intervals of time exclusively to each direction, called time division duplexing (TDD) or ping-pong. Of course, all systems which share the cable must use the same method, and the same duplexing parameters; for TDD this requires all systems to be synchronised.

6.3 Standards

Hitherto, the frequency plan for narrowband ISDN was developed in the international standards bodies, and resulted in a very simple method of control. Simply stated, connection was only permitted using ISDN equipment which conformed to the standard.

It was originally assumed that this approach would serve the higher rate DSL systems too. However, as other nations’ cable configurations differ significantly from those in the UK, different frequency plans are appropriate for them.

Germany’s lines, for example, are almost wholly underground, so RFI is not as significant, and they can permit higher transmitter power levels. Whereas the lines in the Netherlands have much lower spread of lengths in any one cable, so power control is not of such importance. Hence, the international standards include configuration options, and any one nation’s frequency plan will limit the options allowed in that country.

¹⁰ Terminology: ‘upstream’ is the channel towards the exchange; ‘downstream’ is the channel toward the customer.

6.4 The ADSL band

Ongoing work at BT Laboratories is concerned with developing the tools necessary to be able to determine a plan for the spectral management of DSL systems connected to multi-pair access cabling at frequencies up to 1.1 MHz.

The description in this subsection is intended to acquaint the reader with the general principles of developing such a plan and is not to be interpreted as a plan suitable for the UK network.

Assume that the transport technologies to be supported are existing POTS, narrowband ISDN (ISDN-BA), HDSL and new ADSL connections. In each case, one end of the connection is in the exchange, and the other is in the customer's premises¹¹.

Then the frequency plan is designed to permit the existing installed base of equipment, and support a standard form of ADSL [19], while forbidding any more powerful transmissions (which may have an adverse impact on the service quality of existing DSL systems). Furthermore, FDD is assumed for ADSL.

The upstream mask is, for each frequency, the maximum of the spectra produced by ISDN-BA, HDSL and ADSL transmitters. As HDSL has limited reach, in both two-pair and three-pair implementations, three different masks are defined for locations at different ranges from the exchange. Figure 14 is the resulting upstream mask for ranges which three-pair HDSL can reach but two-pair cannot.

The downstream mask is also the maximum of the spectra produced by ISDN-BRA, HDSL and ADSL transmitters. The ADSL bandwidth in this direction is much wider, and dominates the higher frequencies (ISDN dominates the lower frequencies). The resulting mask is shown in Fig 15.

Under this plan the upstream channel is NEXT-limited; since its bandwidth is also available to the downstream channel, receivers at the exchange experience crosstalk from transmitters at the exchange in the same bandwidth. However, the downstream channel has parts which are exclusively reserved for that direction, and this bandwidth offers FEXT-limited capacity.

¹¹ DACS also uses this band, and conforms to this plan. It is effectively an ISDN-BRA connection with the 'customer' end possibly inside BT street furniture.

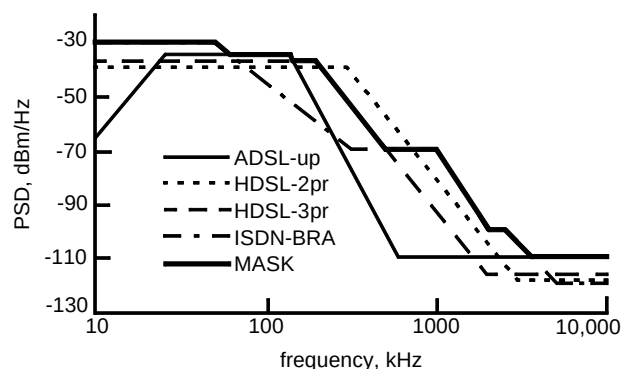


Fig 14 An example upstream mask.

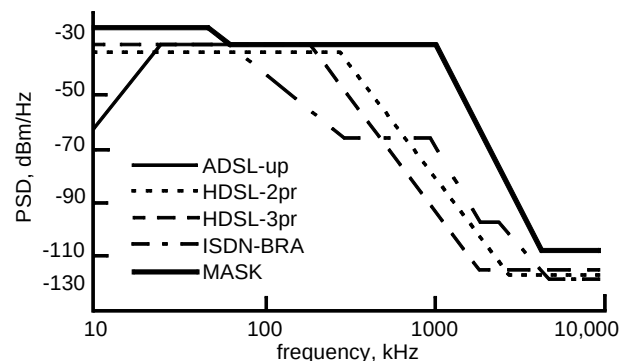


Fig 15 An example downstream mask.

6.5 The VDSL band

Equipment is being developed to use frequency bands above ADSL, which could also work in the same access network. This technology is called VDSL.

It is not yet possible to outline a frequency plan to include it because many technical details have yet to be finalised. For example, the international standards committees are still debating whether such equipment shall use FDD or TDD as the duplexing technique. If TDD is adopted by a network operator, this will necessitate the formulation of a 'time' as well as 'frequency' plan and necessitate an added dimension of complexity to an already complex problem.

However, it is worth noting what is already known about VDSL, so that other parts of the frequency plan leave capacity unassigned.

- VDSL will normally operate over a partial segment of the local access line between the customer premises (NT) and an intermediate node (containing the VDSL LT). In BT terminology the LT would be located at the primary cross-connection point (PCP) — the familiar green cabinet. The PCP will probably connect to the serving exchange via fibre.

Thus, the locations for which the plan will define masks are the customer premises and the cross-connect cabinet. If multiple masks are necessary, range will be defined by distance measured from the intermediate node.

- VDSL will need power back-off in the upstream channel [20], because lines of widely varying length will share cable from the LT (a property of the geography of the UK cable network).

In terms of transmitter masks, power back-off implies a large number of masks; it may be more appropriate to specify the spectrum at the LT receiver.

Note that an ADSL system would be unable to sense the range of the intermediate node, so cannot perform the necessary back-off. Hence an ADSL system may not use the upstream VDSL band, even though either system's upstream transmitter would be sited in the same location.

- Existing equipment should not pollute the VDSL band.

In the past, specifications did not explicitly limit the noise floor at such high frequencies; no use for them was anticipated. Fortunately, measurements show that installed equipment does not actually produce significant power at these frequencies, and future BT procurement specifications will ensure that this continues to be the case.

6.6 The significance of a frequency plan

A frequency plan such as described here has enormous significance because installed equipment which conforms to the plan would be very expensive to adapt to conform to another plan. Thus, a frequency plan will become effectively permanent once it is agreed.

However, the advantages of having a plan far outweigh the drawbacks. For example, the ADSL part of the plan outlined above was written around current business activities, and could become suboptimal for different service aspirations. This is a drawback, but the protection afforded to capacity, even by a somewhat suboptimal plan, is more valuable than an optimal plan which would never be completed.

Producing a plan has a long lead time, partly because it rests on technical details from the international standards bodies (which themselves require a lot of effort and collaboration), and partly because it may have impact beyond an incumbent telco's direct control, e.g. if deregulation were to require open access to the incumbent's wire pairs.

7. Conclusions

The DSL technologies outlined in this paper are still evolving. Each new generation brings improvements in functionality, performance and levels of integration. This trend of technology development and innovation looks set to continue, in the same way that voice-band modems evolved to exploit the capacity of the voice channel more efficiently.

Since the key limitation to DSL capacity is crosstalk, BT has invested considerable effort in understanding the crosstalk environment of its network. This measurement and modelling effort is enabling the real-world performance of new DSL systems to be quantified, which in turn helps BT to work with its suppliers to optimise system designs and international standards. It also facilitates more accurate performance predictions and equipment deployment rules.

A critical aspect of the deployment rules for DSL involve ensuring that new generations of DSL and existing legacy copper transmission systems do not mutually interfere. There are now so many variants of DSL with extremely flexible configuration options, that it would be easy for a telco with limited expertise in this area to accidentally deploy a system that precludes future evolution options, and artificially limits the potential performance of their network. Worse still, it could degrade the performance of existing systems and even cause service interruption to customers whose traffic is carried on such systems.

Hence it is vital for BT, and the UK as a whole, to have world class expertise in this area. Consequently, BT is expending significant effort in understanding the spectral compatibility issues for the copper network and developing a frequency plan to ensure that the universal broadband access potential of the existing access network is fully realised.

The results of this effort are already bearing fruit. For example, together with telco colleagues in FSAN [21], notable successes have been achieved in modifying the spectral masks within existing DSL standards to ensure that standards-compliant systems are fit for use in the BT network. Vendor acceptance of such changes would be impossible without technical credibility, and telco consensus.

Standards are necessary, but not sufficient. It is still possible to use standards-compliant equipment in configurations that reduce available capacity, or impact the performance of existing systems. Hence, work is continuing to ensure that BT develops planning rules which maximise the overall capability of the copper access network.

Protecting the potential capacity of the local access network requires co-ordinated management of all the

transmission systems connected to it, and limitation of their individual spectral power transmission levels. While any one plan is unlikely to be optimal for all possible uses of the network, it is very important that such a plan be developed and is made mandatory. Failure to do so would deprive all potential users of most of the potential capacity — a lose-lose scenario.

This work is all the more vital because many of the technical decisions that feed into equipment procurement specifications, international standards and planning rules can only be made once. As soon as access equipment is deployed it is very expensive to recover it in favour of 'the next best thing'.

In summary, the telco installed copper pair network presents a challenging and hostile environment for high-speed transmission. Telcos can leverage this asset to release the potential for competitive broadband services using DSL technologies. However, realisation of the megabit capacity of the existing network is critically dependent on understanding and controlling the crosstalk environment via the use of a mandatory frequency plan to ensure spectral compatibility of new and legacy transmission systems.

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